HÖGSKOLAN DALARNA

HÖGSKOLAN DALARNA

Test Plan

System Maintenance and Test of IT-Systems

Course Code: GIK2PB

Authors: Ali Adrian Nasrat, Vehid Sumic, Anna Backolars, Nengxi Peng

Group: 5

HÖGSKOLAN
DALARNA

Table of contents

1 Introduction

Background

This test plan outlines how our project group will evaluate the accessibility and usability of Dalarna University's English website (https://www.du.se/en) in the course *System Maintenance and Test of IT-Systems (GIK2PB)*. The evaluation is driven by the shared user story:

"As an employee, I want to know if I can get a wellness allowance to buy an exercise app."

To capture diverse needs, each group member tests from one persona's perspective:

- **P001 – Visual impairment (blind):** screen-reader access, semantic structure, alternative text.
- **P002 – Active mobile user:** mobile responsiveness, reflow, orientation, tap accuracy.
- **P003 – Motor difficulties:** keyboard operability, no keyboard traps, gesture-free flows, target size.
- **P004 – Inexperienced computer/mobile user:** clarity of labels/instructions, consistent navigation, error prevention.

We apply a **dynamic, black-box, non-functional** testing approach focused on accessibility. We follow **WCAG 2.1 Level AA** as the primary conformance target, and we use both **manual techniques** and **automated tools** (e.g., WAVE, ANDI) to identify barriers. In line with established practice, our test artifacts (test plan, test cases, logs, error reports, summary) provide traceability from objectives to evidence (Hambling, 2010).

Why Testing Is Important

Accessibility testing ensures that users with varying abilities can **perceive, understand, navigate, and interact** with the university's digital content. (Hambling, 2010)

Concretely, the risks we aim to control include:

- **Exclusion risks:** Blind users cannot reach or interpret content if landmarks, headings, and alternative texts are missing (WCAG 1.1.1, 1.3.1, 2.4.x).
- **Mobile usability risks:** Content that doesn't reflow or maintain readable tap targets can frustrate active mobile users (WCAG 1.4.10, 2.5.5).
- **Motor-control risks:** Small or tightly spaced controls, mandatory drag/precision gestures, or keyboard traps can block users with motor difficulties (WCAG 2.1.1, 2.1.2, 2.5.1, 2.5.2, 2.5.5).

By systematically testing against **WCAG 2.1 AA**, we reduce these risks, increase confidence in the site's accessibility posture, and produce actionable findings that align with recognized standards rather than subjective opinion.

How It Will Help the School

Improving accessibility has **practical, educational, and organizational** benefits for Dalarna University:

- **Practical outcomes for staff and students**

  Staff can independently locate information such as the **Health and Wellness Allowance** policy without assistance, saving time for both employees and support teams. Clearer navigation and compliant components decrease help-desk load and repeated inquiries.

- **Compliance and reputation**

  Demonstrable alignment with **WCAG 2.1 Level AA** strengthens compliance with Swedish/EU accessibility expectations, lowers legal/regulatory exposure, and reflects the university's public commitment to **digital inclusion**.

- **Maintainability and continuous improvement**

  The defects and recommendations we document (e.g., size adjustments on key targets, keyboard focus fixes, heading structure corrections) are **specific, testable, and repeatable**, making them efficient to prioritize and verify in future releases. This supports a sustainable maintenance cycle and better long-term quality (Hambling, 2010).

- **Pedagogical value**

  As an academic institution, Dalarna University benefits from modeling best practices in accessible design and testing. The findings from this plan can inform internal guidelines, developer checklists, and training for editors and web teams.

In short, this testing helps ensure that **all employees regardless of device, experience, or ability, can complete critical tasks on the university website**, while giving the school a clear roadmap for enhancements.

1.1 Scope

1.1.1 In Scope

This accessibility and usability evaluation focuses on the **Dalarna University English website** (https://www.du.se/en) and aims to determine whether employees with different abilities can independently locate and understand information about the **Wellness Allowance**, particularly in relation to purchasing an **exercise app**.

Testing covers how four personas, **P001 (Visual Impairment)**, **P002 (Active Mobile User)**, **P003 (Motor Difficulties)**, and **P004 (Inexperienced Computer/Mobile User)** interact with the site.

All tests are performed in accordance with **WCAG 2.1 Level AA**, which defines internationally accepted standards for accessible web content.

The following areas are included within scope:

- **Search functionality:**

  Evaluate how easily the search feature can be used to locate the *Health and Wellness Allowance* page using typical and alternative search terms (e.g., *allowance*, *wellness*, *exercise app*).

  Tests will be conducted with a keyboard, mouse, touch, and standard input methods across devices.

- **Navigation and information retrieval:**

  Assess how efficiently users can move through menus, headings, and search results to reach the correct information page.

- **Content readability and visibility:**

  Verify that text, colors, and contrasts remain legible, including when zoomed up to 200 %.

  Ensure focus indicators and link labels are visible and descriptive.

- **Interactive elements:**

  Examine buttons, links, and menus to ensure sufficient target size, proper spacing, and accessibility via keyboard, mouse, and touch input.

  Confirm that no keyboard traps or fine-motor gestures are required.

- **Responsiveness and mobile usability:**

  Validate that all key content reflows correctly and that interactive elements function properly on both desktop and mobile devices (Chrome / Edge / Safari / iOS / Android).

HÖGSKOLAN DALARNA

- **Usability for inexperienced users:**

  Confirm that the site uses consistent navigation, intuitive structure, and clearly labeled elements so that less experienced users can complete the task without guidance.

- **Compliance with relevant WCAG 2.1 criteria**, including but not limited to:
  - **1.4.1 Use of Color**
  - **1.4.3 Contrast (Minimum)**
  - **1.4.4 Resize Text**
  - **1.4.10 Reflow**
  - **2.1.1 Keyboard Accessibility**
  - **2.1.2 No Keyboard Trap**
  - **2.4.3 Focus Order & 2.4.7 Focus Visible**
  - **2.5.1 Pointer Gestures & 2.5.2 Pointer Cancellation**
  - **2.5.5 Target Size**
  - **3.2.3 Consistent Navigation & 3.3.2 Labels or Instructions**

Both **manual and automated testing** will be used:

- *Manual testing* simulates realistic scenarios for each persona using keyboards, screen readers, and touch input.
- *Automated tools* such as **WAVE** and **ANDI** will detect structural and semantic accessibility issues.

  Testing will be performed on multiple devices to ensure a consistent experience for all personas.

1.1.2 Out of Scope

The following aspects fall outside the scope of this lab:

- **Non-accessibility-related testing**, such as backend processes, server performance, or database integrations.
- **Content accuracy and translation** of the wellness-allowance text (e.g., Swedish/English consistency).
- **Unrelated website areas**, including course catalogues, research sections, and student login portals.
- **Personas not covered** in this project (e.g., hearing or cognitive impairments beyond the defined four).
- **Browser compatibility** beyond Chrome, Edge, and Safari, unless accessibility defects are directly observed in other browsers.
- **Third-party content or embedded files** not managed by Dalarna University.

This unified scope ensures that all four personas are represented under a single, comprehensive framework. It provides a clear boundary for what the group will test and guarantees consistency in evaluating how well the Dalarna University website meets **WCAG 2.1 Level AA** accessibility standards.

1.2 Quality Objective for persona with Motor Difficulties

3.1 Quality Objective (Shared Across Personas)

The overall quality objective of this accessibility and usability test is to ensure that the **Dalarna University website** (https://www.du.se/en) provides an **inclusive, barrier-free, and user-centered experience** for all employees regardless of visual, motor, situational, or digital proficiency limitations.

The goal is that each user can independently locate, read, and understand information about the **Wellness Allowance**, including whether it can be used to purchase an **exercise app**, using both desktop and mobile devices.

This objective aligns with the **Web Content Accessibility Guidelines (WCAG) 2.1 Level AA**, which represent internationally recognized best practices for accessible digital content. The testing ensures that the site is not only compliant with technical standards but also *fit for purpose* (Hambling, 2010), meeting real users' needs in their respective contexts.

The evaluation focuses on four main accessibility dimensions, represented by distinct personas:

Persona	Accessibility Focus	Quality Criteria
P001 – Visual Impairment (Blind / Low Vision)	Ensuring that text, images, and interface elements are perceivable and operable through visual enhancements or assistive technologies.	- Sufficient color contrast (WCAG 1.4.3)- Readability at 200% zoom (WCAG 1.4.4)- Clear focus indicators (WCAG 2.4.7)- Descriptive headings and link labels
P002 – Active Mobile User	Ensuring the website remains responsive, clear, and operable through touch interaction on smaller screens.	- Reflow and orientation flexibility (WCAG 1.4.10, 1.3.4)- Proper target size and spacing (WCAG 2.5.5)- Accessible buttons and menus on mobile
P003 – Motor Difficulties	Ensuring that all functionalities are operable without fine-motor precision or multi-finger gestures.	- Full keyboard accessibility (WCAG 2.1.1)- No keyboard traps (WCAG 2.1.2)- Pointer gesture alternatives and cancellation (WCAG 2.5.1–2.5.2)- Large, well-spaced interactive targets (WCAG 2.5.5)
P004 – Inexperienced Computer / Mobile User	Ensuring intuitive navigation, readable layout, and consistency for users with limited technical skills.	- Consistent navigation (WCAG 3.2.3)- Clear labels and instructions (WCAG 3.3.2)- Readable structure and intuitive flow (WCAG 1.4.10)

By meeting these quality objectives, Dalarna University strengthens its commitment to **digital inclusion**, **equal access**, and **sustainable system maintenance**.

This ensures that the website serves all employees effectively while fulfilling Swedish and EU accessibility standards.

3.2 Roles and Responsibilities

Defining clear roles and responsibilities ensures structure, accountability, and traceability throughout the testing process (Hambling, 2010).

Each tester in the group represents a specific persona and focuses on unique accessibility aspects of the university's website. Collaboration and knowledge sharing between members guarantee full coverage across accessibility dimensions.

Persona / Role	Responsibilities	Focus & Tools
P001 – Tester for the Visual Impairment Persona	- Conduct tests simulating low-vision conditions (e.g., magnification, high contrast). - Verify text readability, focus indicators, and descriptive alt text. - Identify barriers for screen-magnifier and color-contrast users. - Report findings on visual clarity and structural hierarchy.	WCAG 1.4.1–1.4.4, 2.4.7; Tools: **WAVE**, **ANDI**, browser zoom (200%), contrast checker.
P002 – Tester for the Active Mobile User Persona	- Evaluate navigation and content readability on smartphones (iOS and Android). - Check reflow, tap target size, and orientation responsiveness. - Identify usability issues related to touch interaction. - Record findings and screenshots across mobile browsers.	WCAG 1.4.10, 1.3.4, 2.5.5; Tools: **WAVE**, **ANDI**, Chrome/Safari Mobile.
P003 – Tester for the Motor Difficulties Persona (Adrian)	- Design and execute accessibility tests focusing on keyboard-only navigation and pointer alternatives. - Evaluate operability of search, menus, and buttons without fine-motor gestures. - Use **WAVE** and **ANDI** to detect WCAG 2.1.1–2.5.5 violations. - Document both functional (e.g., clickable areas) and usability findings (e.g., layout spacing).	WCAG 2.1.1, 2.1.2, 2.5.1–2.5.5; Tools: **Keyboard-only input**, **ANDI**, **WAVE**.
P004 – Tester for the Inexperienced User Persona	- Evaluate the clarity and intuitiveness of navigation and content structure. - Test the search function with multiple terms and verify results consistency. - Assess text readability and ease of use on desktop and mobile. - Document confusing elements or unclear wording.	WCAG 3.2.3, 3.3.2, 3.3.6; Tools: **WAVE**, **ANDI**, Chrome Desktop, Safari Mobile.

3.3 Collaborative Responsibilities (Group-Wide)

- **Design and Planning:** All members contribute to defining the scope, objectives, and success criteria for the test plan.
- **Execution:** Each member executes at least four test cases for their persona, ensuring device and browser diversity.
- **Documentation:** All findings are recorded in shared **Test Logs** and **Error Reports**, including screenshots, WCAG references, and severity levels.
- **Review and Discussion:** Weekly group discussions are held to review discovered defects, assign priorities through **bug triage**, and determine next steps.
- **Reporting:** The team collectively finalizes the **Test Report Summary**, integrating results from all personas for a holistic evaluation.

By defining these responsibilities and aligning them with the group’s shared quality objectives, the testing process remains structured, transparent, and comprehensive. Each persona’s unique findings strengthen the collective understanding of accessibility barriers and improvement opportunities.

2 Methodology

2.1 Overview

he testing will be conducted as an **accessibility and usability evaluation** of the Dalarna University website (https://www.du.se/en), focusing on identifying barriers that may prevent users with various accessibility needs from locating and understanding information about the **wellness allowance** for employees.

Each group member represents a distinct **persona** with different accessibility challenges:

- **P001 – Visual impairment (blind):** evaluates compatibility with screen readers and checks that all images, forms, and structural elements include accessible labels and alternative text.
- **P002 – Active mobile user** verifies that the website functions smoothly on mobile devices, ensuring that layout, tap targets, and orientation remain usable on the go.
- **P003 – Motor difficulties:** tests the site's ability to be fully operated through keyboard-only input or simple pointer actions, avoiding fine motor or gesture-dependent interactions.
- **P004 – Inexperienced computer/mobile user** focuses on overall clarity, intuitive navigation, and whether essential information can be found without prior technical knowledge.

The testing approach follows a **dynamic, black-box, non-functional testing method** (Hambling, 2010), meaning that the internal code structure of the website will not be examined. Instead, the evaluation focuses on how the system behaves externally specifically how users interact with its interface, navigation, and content. This aligns with accessibility testing principles, which aim to assess how effectively a system supports real user interaction rather than how it is implemented internally.

Testing will be performed using both **manual and automated techniques** to ensure thorough coverage across personas:

- **Manual testing** simulates real-world user behavior for each persona:
  - For **P001**, navigating the site with a **screen reader** to verify proper reading order and alt-text.
  - For **P002**, accessing the site on **mobile browsers** to test reflow, tap size, and screen rotation.
  - For **P003**, using only **keyboard navigation** (Tab, Shift+Tab, Enter) to test focus visibility, pointer gestures, and target spacing.
  - For **P004**, performing typical browsing actions to assess how intuitive and self-explanatory the site is for less experienced users.

- **Automated testing** will be conducted using tools such as **WAVE** (Web Accessibility Evaluation Tool) and **ANDI** (Accessible Name & Description Inspector) to identify missing accessibility attributes, improper heading hierarchy, or violations of **WCAG 2.1 Level AA** success criteria.

Testing will take place on both **desktop and mobile platforms**, in accordance with lab requirements ensuring that all user stories can be completed on any device.

Each persona will focus on relevant WCAG 2.1 success criteria, for example:

- **1.1.1 Non-text Content** – providing alt text for images and icons (P001).
- **1.4.10 Reflow & 2.5.5 Target Size** – maintaining readability and sufficient tap area on small screens (P002).
- **2.1.1 Keyboard Accessibility & 2.5.1 Pointer Gestures** – ensuring complete keyboard operation and gesture alternatives (P003).
- **3.2.3 Consistent Navigation & 3.3.2 Labels or Instructions** – providing predictable, easy-to-understand navigation and clear feedback (P004).

The testing process will be **iterative**, involving the identification of issues, documentation of defects, and group discussions about potential improvements. All findings will be recorded in a **Test Log** and summarized in an **Error Report**, which together form part of the deliverables for Lab 1 and Lab 2.

As Hamilton (n.d.-a) explains, accessibility testing ensures that software systems meet both **usability** and **compliance** standards by combining automated analysis with human judgment. This hybrid approach allows a comprehensive understanding of how different users experience the website — emphasizing not only technical compliance but genuine **user inclusiveness**.

2.2 Test Levels

Accessibility testing of the Dalarna University website (https://www.du.se/en) is conducted according to the **Web Content Accessibility Guidelines (WCAG) 2.1**, which define three conformance levels:

- **Level A** – the minimum level, covering essential accessibility features that must be present for basic access.
- **Level AA** – the mid-range level, addressing the most common and impactful barriers for users with disabilities.
- **Level AAA** – the highest level, representing enhanced accessibility and advanced best practices.

For this lab, testing is performed primarily against **WCAG 2.1 Level AA**, as required by the course and by current Swedish and EU accessibility legislation.

Level AA ensures that the website can be used effectively by a broad range of users including those with visual impairments, motor difficulties, or limited technical experience across both desktop and mobile devices.

The testing process follows a **black-box, usability-oriented approach**, where testers interact with the website without reviewing its source code.

As Hambling (2010) notes, this form of testing focuses on *how well* a product meets user needs and usability expectations, particularly in the context of accessibility and inclusive design.

Personas and Relevant WCAG 2.1 Criteria

Each group member applies WCAG 2.1 criteria most relevant to their assigned persona, as outlined below.

Persona ID	Persona Name	Key WCAG 2.1 Criteria	Level	Description
P001	**Visual Impairment (Blind)**	1.1.1 Non-text Content	A	All non-text elements (images, icons) must have meaningful alternative text.
		1.3.1 Info and Relationships	A	Content structure (headings, lists, tables) must convey relationships properly to screen readers.
		2.4.1 Bypass Blocks	A	Provide mechanisms to skip repetitive content (e.g., “Skip to main content”).
		2.4.6 Headings and Labels	AA	Headings and labels must describe topic or purpose clearly.
P002	**Active Mobile User**	1.4.10 Reflow	AA	Content must reflow without loss of information when zoomed or viewed in portrait/landscape orientation.
		2.5.5 Target Size	AAA	Touch targets must be large enough and well-spaced for single-handed use.
		1.3.4 Orientation	AA	Website must support both portrait and landscape modes.
		2.5.1 Pointer Gestures	A	Complex gestures (e.g., drag, pinch) must not be required.
P003	**Motor Difficulties**	2.1.1Keyboard	A	All functionalities must be operable through a keyboard interface.
		2.1.2 No Keyboard Trap	A	Keyboard users must not become trapped in any element or region.
		2.5.1 Pointer Gestures	A	Avoid reliance on gestures requiring fine motor control.
		2.5.2 Pointer Cancellation	A	Users must be able to cancel unintended taps or clicks easily.
		2.5.5 Target Size	AAA	Interactive elements must have sufficient size and spacing.
P004	**Inexperienced Computer/Mobile User**	3.2.3 Consistent Navigation	AA	Navigation mechanisms must be consistent across pages.
		3.3.2 Labels or Instructions	A	Provide clear labels and instructions for forms and inputs.
		3.3.6 Error Prevention (All)	AA	Forms should help prevent input errors or provide confirmation before submission.
		2.4.3 Focus Order	A	Focus must move in a logical, predictable sequence.

Testing Focus

Each persona's testing will verify that these criteria are satisfied on both **desktop** and **mobile** devices when performing the shared user story:

"As an employee, I want to know if I can get a wellness allowance to buy an exercise app."

Testing against **Level AA** ensures that the core accessibility requirements for users with disabilities or limited digital experience are fulfilled.

Criteria from **Level AAA** (e.g., Target Size 2.5.5) will be considered as enhancements to further improve usability and inclusiveness.

2.3 Bug Triage

The **bug triage process** ensures that all identified accessibility issues are reviewed, prioritized, and documented systematically. It helps the group decide which defects are most critical to users and which should be resolved first. According to Hambling (2010), effective triage is essential to maintaining quality control and optimizing testing effort particularly when multiple testers and user perspectives are involved.

In this project, bug triage involves all four testers representing different personas:

- **P001 – Visual Impairment (Blind)**
- **P002 – Active Mobile User**
- **P003 – Motor Difficulties**
- **P004 – Inexperienced Computer/Mobile User**

Each tester logs their discovered defects in a shared **Test Log**, including screenshots, WCAG references, and device/browser details. During group review meetings, the team discusses severity, assigns priority levels, and determines whether retesting is required.

Bug Prioritization Levels

Priority Level	Description	Example (Across Personas)
Critical	Prevents the user from completing the core task or accessing key information.	- P001: Screen reader cannot access main navigation or page headings. - P002: Website fails to load or respond on mobile devices. - P003: Search field cannot be reached via keyboard. - P004: Essential buttons (e.g., “Search”) lack visible text labels.
High	Significantly impacts accessibility but can be temporarily worked around.	- P001: Some images missing alt text. - P002: Tap targets too close together. - P003: Buttons require precise mouse control. - P004: Inconsistent page layout or unclear instructions.
Medium	Reduces usability but does not fully block access.	- P001: Decorative images not marked properly. - P002: Minor layout overlaps on mobile. - P003: Keyboard focus outline inconsistent. - P004: Navigation labels vary slightly between pages.
Low	Minor cosmetic issues with limited accessibility impact.	- P001: Slight heading hierarchy inconsistency. - P002: Font color contrast slightly below optimal. - P003: Focus indicator styling could be improved. - P004: Redundant tooltip text on icons.

Bug Reporting and Tracking Process

1. **Identification and Documentation**
    - Each tester records issues found during testing using the *Test Log*.
    - Each entry includes a unique ID, date, device, persona, description, WCAG criterion violated, and severity.
2. **Review and Prioritization**
    - The group meets to review all reported issues and agree on a final **priority level** based on user impact.
    - Testers verify whether similar defects appear across multiple personas.
3. **Assignment and Tracking**
    - The group assigns responsibility for retesting once fixes or workarounds are implemented.
    - Issues are tracked collectively to ensure accountability.
4. **Verification and Closure**
    - After a defect is fixed or mitigated, the assigned tester re-runs the relevant test case.
    - The issue is marked as **Closed** when compliance with WCAG criteria is confirmed.

Example of Cross-Persona Defect Collaboration

A typical scenario might involve one defect impacting multiple personas:

- **Example:** The "Search" button on the homepage lacks a visible label (<button> has only an icon).
    - *P001 (Blind):* Screen reader fails to announce the button's purpose → **Critical**.
    - *P003 (Motor Difficulties):* Small icon difficult to select → **High**.
    - *P004 (Inexperienced User):* Icon meaning unclear → **High**.
    - *P002 (Mobile):* Icon visible and tappable but may lack alternative text → **Medium**.

This collaborative triage ensures that issues affecting multiple accessibility areas are recognized and prioritized effectively.

Tools and Documentation

- **WAVE** and **ANDI** reports will be used as objective references when confirming WCAG violations.
- A shared **Google Sheet** or **Excel Test Log** will store issue IDs, descriptions, WCAG references, and status updates.
- Screenshots and URLs will accompany all Critical and High-level defects for evidence and reproducibility.

2.4 Suspension Criteria and Resumption Requirements

The **suspension criteria** define conditions that temporarily halt testing when valid progress or accurate evaluation cannot continue.

The **resumption requirements** specify when testing may safely resume.

According to Hambling (2010), controlled suspension maintains reliability by avoiding false or incomplete results when test environments, tools, or prerequisites fail.

Accessibility testing of the Dalarna University website (https://www.du.se/en) may be suspended if:

- The website, its sub-pages, or key functions (e.g., search) are unavailable.
- Automated tools (**WAVE**, **ANDI**) produce incomplete or inconsistent results.
- Critical browser or device errors occur that prevent test execution.
- Any tester encounters an accessibility barrier that blocks progress within their persona context (e.g., keyboard trap, unreadable screen-reader output).

Testing will resume once the blocking issue is corrected, verified, or a suitable workaround is in place.

WCAG 2.1 Checklist and Suspension Triggers per Persona

Persona ID	Relevant WCAG Criterion	Level	Suspension Condition	Resumption Requirement
P001 – Visual Impairment (Blind)	1.1.1 Non-text Content / 2.4.1 Bypass Blocks / 2.4.6 Headings & Labels	A / AA	If the screen reader cannot detect or read main headings, links, or labels; or if the page lacks a logical reading order.	Resume after alt texts and heading hierarchy are verified, and screen reader navigation works correctly.
P002 – Active Mobile User	1.4.10 Reflow / 2.5.5 Target Size / 1.3.4 Orientation	AA / AAA	If mobile layout breaks (content overlaps or becomes unreachable) or touch targets are too small to activate.	Resume once responsive layout and adequate target spacing are restored across orientations.
P003 – Motor Difficulties	2.1.1 Keyboard / 2.1.2 No Keyboard Trap / 2.5.1 Pointer Gestures / 2.5.2 Pointer Cancellation	A	If keyboard focus is lost, trapped, or gesture-based controls cannot be performed without fine motor movement.	Resume after keyboard navigation is restored and gesture-free alternatives function as intended.

Persona ID	Relevant WCAG Criterion	Level	Suspension Condition	Resumption Requirement
P004 – Inexperienced Computer / Mobile User Inexperienced Computer / Mobile User	3.2.3 Consistent Navigation / 3.3.2 Labels or Instructions / 3.3.6 Error Prevention	A / AA	If page instructions are missing, inconsistent, or error messages make continuation unclear.	Resume after navigation and labels are standardized and guidance restored for task completion.

Example of Suspension Scenario

If the *Search* function on https://www.du.se/en becomes inaccessible due to a temporary script error:

- **P001** may suspend because the screen reader cannot locate the search field.
- **P003** may suspend because keyboard focus skips the input.
- **P002 / P004** continue other tests not dependent on search.

  Testing resumes when the search field regains full accessibility and consistency across devices.

Process

1. **Detection** – Tester notes the blocking issue, logs it as "Suspended" in the Test Log with persona ID and WCAG criterion.
2. **Communication** – All group members are informed via shared document or meeting.
3. **Verification** – After fix/workaround, tester retests to confirm restoration of functionality.
4. **Resumption** – Testing proceeds from the last successfully executed step; log entry updated to "Resumed."

This systematic suspension/resumption process ensures the test results remain consistent, traceable, and valid across all personas.

2.5 Test Completeness

Testing will be considered **complete** when all planned accessibility test cases for the four personas have been executed and evaluated according to the scope and quality objectives of this plan.

The purpose of this phase is to verify that https://www.du.se/en enables all users—regardless of visual, motor, situational, or cognitive limitations—to locate and read information about the **Health and Wellness Allowance** independently.

According to Hambling (2010), test completeness is achieved when the defined coverage criteria have been met, all critical defects have been addressed, and the remaining issues do not prevent the system from fulfilling its intended purpose.

For this group project, completeness encompasses both **technical compliance with WCAG 2.1 Level AA** and **practical usability** for the assigned personas.

Completion Criteria

Criterion	Description
Execution of All Test Cases	Each persona (P001–P004) has executed a minimum of four test cases using both manual and automated methods on desktop and mobile.
Verification of WCAG 2.1 Conformance	All applicable Level A and AA success criteria have been evaluated for compliance, including alt-text, keyboard control, reflow, target size, and navigation consistency.
Resolution of Critical and High Defects	All issues rated Critical or High during the bug-triage process have been fixed or documented with approved workarounds.
Cross-Persona Validation	Findings have been reviewed collectively to confirm that the website is accessible across multiple user types and devices.
Group Review and Approval	The entire team has reviewed, discussed, and approved the consolidated results before submission.

Exit Criteria

Testing can be formally closed when:

- 100 % of the planned test cases have been executed,
- At least **90 % of test cases pass**,
- No **Critical** or **High** defects remain unresolved,
- Remaining **Medium/Low** issues are documented with minimal user impact, and
- All test documentation (plan, cases, logs, and reports) has been finalized and reviewed.

Persona-Specific Indicators of Completion

Persona ID	Indicator of Successful Completion
P001 – Visual Impairment (Blind)	Screen reader reads page titles, headings, and link text in a logical order; all non-text content has descriptive alternatives.
P002 – Active Mobile User	The website maintains correct layout, readability, and tap target spacing in both portrait and landscape orientations.
P003 – Motor Difficulties	All navigation, forms, and controls are operable with keyboard-only input or simple pointer actions; no fine-motor gestures are required.
P004 – Inexperienced Computer / Mobile User	Menus, buttons, and instructions are self-explanatory; tasks can be completed without prior technical knowledge.

Success Definition

Testing is deemed successful when the website meets or exceeds WCAG 2.1 Level AA compliance, and all personas can perform the core task **finding and understanding the wellness-allowance information** without external help.

Achieving this state confirms that Dalarna University's website is *fit for purpose* (Hambling, 2010) and aligns with Swedish and EU accessibility regulations.

HÖGSKOLAN
DALARNA

3 Test Deliverables

The purpose of test deliverables is to document each stage of the testing process—from planning through execution to evaluation in a structured and verifiable way.

According to Hambling (2010), comprehensive test documentation ensures **traceability**, **transparency**, and **accountability** throughout the testing lifecycle.

For this project, all deliverables are collaborative and reflect results gathered across the four personas:

- **P001 – Visual Impairment (Blind)**
- **P002 – Active Mobile User**
- **P003 – Motor Difficulties**
- **P004 – Inexperienced Computer / Mobile User**

All deliverables will be compiled and submitted together as part of the combined **Lab 1 + Lab 2** submission on Canvas.

3.1 Primary of Deliverables

Deliverable	Description	Purpose / Importance	Responsible Party
Test Plan	The current document outlining objectives, scope, personas, methodology, and expected outcomes.	Defines the testing framework, ensures consistent goals, and provides an overview of the group's accessibility approach.	Entire group
Test Cases	A minimum of four test cases per persona, created using the official course template.	Provide detailed, repeatable procedures for verifying accessibility and usability according to WCAG 2.1 Level AA.	Each group member (P001–P004)
Test Log	A record of every test execution, including date, tester, device, results, and defects discovered.	Tracks progress, provides evidence of coverage, and supports later analysis and retesting.	All testers collaboratively
Error Reports (Defect Reports)	Structured documentation describing issues found, severity levels, and affected WCAG criteria.	Enables prioritization and resolution of accessibility barriers following the bug triage process.	All testers collaboratively
Test Report (Summary)	A high-level summary of outcomes, including pass/fail ratios, key findings, and improvement recommendations.	Provides management-level insight into the overall accessibility status and test effectiveness.	Group coordinator or appointed editor

3.2 Supporting Deliverables

Deliverable	Description / Usage
WAVE and ANDI Reports	Automated tool outputs verifying compliance with WCAG 2.1. Used to support manual findings.
Screenshots and Evidence	Visual documentation of test steps, defects, and corrected results. Strengthens traceability and reproducibility.
Meeting Notes	Brief notes from group discussions, bug triage sessions, and retesting coordination meetings.
Accessibility Improvement Suggestions	Persona-based recommendations (e.g., larger buttons, clearer labels, improved alt-text) collected from testers after analysis.

3.3 Integration of Persona Deliverables

All deliverables are integrated into a single package to reflect collaboration among the four testers:

- **P001 (Blind):** Screen-reader evaluations, alt-text verifications, logical structure mapping.
- **P002 (Mobile):** Responsiveness and orientation testing results.
- **P003 (Motor Difficulties):** Keyboard navigation, pointer gesture, and target-size results.
- **P004 (Inexperienced User):** Usability and comprehension results regarding labeling and instruction clarity.

Each persona's results are merged into the **Test Report Summary**, ensuring that all accessibility perspectives are represented collectively.

3.4 Deliverable Quality Control

Before submission, the group will:

1. Review all deliverables for completeness and consistency.
2. Validate that every test case links to at least one WCAG 2.1 criterion.
3. Ensure all defect reports include severity, description, and evidence.
4. Approve the final compiled report collectively.

This structured documentation process guarantees a reliable and auditable outcome that accurately represents the accessibility status of Dalarna University's website.

---

4 Resource & Environment Needs

To ensure consistency, reliability, and repeatability of test results, testing will be conducted within a controlled environment that accurately represents how real users with different abilities interact with the Dalarna University website (https://www.du.se/en).

According to Hambling (2010), defining and maintaining a stable test environment is essential for reproducibility and comparability of results across testers, devices, and test iterations.

Testing will combine **manual evaluations** and **automated analysis**, performed on both **desktop** and **mobile** platforms. Each persona uses specific tools and accessibility settings relevant to their characteristics.

4.1 Testing Tools and Resources

The following testing tools will be used to evaluate accessibility and usability for users with **motor difficulties**:

WAVE (Web Accessibility Evaluation Tool)	All personas	Automated accessibility validation.	Identifies missing alt text, contrast issues, and other WCAG 2.1 violations.
ANDI (Accessible Name & Description Inspector)	All personas	Inspects HTML elements, ARIA attributes, and semantic structure.	Confirms that interactive elements are correctly labeled and accessible.
Screen Reader (NVDA / VoiceOver)	P001 – Visual Impairment	Reads out all text, links, and navigation landmarks.	Validates logical reading order, labeling, and semantic structure.
Mobile Browser (Chrome / Safari)	P002 – Active Mobile User	Tests responsiveness and tap target size.	Confirms accessibility on mobile platforms and orientation adaptability.
Keyboard Navigation (Tab, Shift+Tab, Enter)	P003 – Motor Difficulties	Tests operability without mouse input.	Simulates limited motor control by relying on keyboard alone.
Standard Browser (Chrome / Edge)	P004 – Inexperienced User	Conducts basic usability and comprehension testing.	Evaluates clarity, labeling, and error prevention for less experienced users.
Screen Magnifier / Browser Zoom	P001 & P003	Tests readability and scalability at 125–150 %.	Ensures that text and controls remain usable when enlarged.
Internet Connection	All personas	Stable network connection.	Required for loading website content.

4.2 Test Environment Configuration

Testing will take place on a combination of desktop and mobile environments to verify accessibility across platforms.

Environment Component	Specification / Example	Purpose
Operating Systems	macOS 14 (Sonoma), Windows 11, Android 14, iOS 18	Validate accessibility across multiple OSs and device types.
Web Browsers	Google Chrome (Version 129+), Microsoft Edge, Safari (iOS), Firefox Desktop	Ensure compatibility with the most common browsers.
Devices	Laptop (keyboard + trackpad), smartphone (iPhone / Android), external keyboard	Enable both desktop and mobile accessibility testing.
Accessibility Settings	High-contrast mode, zoom, voice feedback, keyboard navigation	Simulate realistic usage conditions for accessibility testing.
Network Requirements	Stable broadband / Wi-Fi connection	Guarantee site availability and reliable tool performance.

4.3 Persona-Specific Environments

Persona ID	Testing Setup	Focus Area / Tools
P001 – Visual Impairment (Blind)	Desktop or laptop with screen reader (NVDA on Windows / Voiceover on macOS).	Verifies text alternatives, structure, and navigation readability.
P002 – Active Mobile User	Smartphone with Chrome / Safari; portrait and landscape testing.	Checks reflow, orientation, and touch-target spacing.
P003 – Motor Difficulties	Desktop/laptop using only keyboard; optional adaptive mouse or switch device.	Ensures all controls are operable via keyboard or simple pointer input.
P004 – Inexperienced Computer / Mobile User	Desktop or mobile device using standard browser configuration.	Tests clarity of menus, labels, and instructions; overall ease of navigation.

4.4 Environment Validation

Before testing begins, all testers will verify that:

1. Tools such as WAVE and ANDI load correctly in each browser.
2. Accessibility settings (e.g., keyboard shortcuts, screen readers, zoom) function as expected.
3. The university's website is accessible from both mobile and desktop devices.
4. All test data and screenshots can be stored in a shared folder (e.g., Google Drive / OneDrive) for group review.

4.5 Environment Maintenance

To maintain a consistent environment throughout testing:

- Browsers will not be upgraded mid-testing phase unless necessary for security reasons.
- Each tester will record browser version, OS version, and device model in the **Test Log**.
- Accessibility tools will be configured identically across group members to ensure comparable results.

5 Terms/Acronyms

Term / Acronym	Definition
Accessibility Testing	A non-functional software-testing process that verifies whether a system can be used effectively by people with disabilities, following the *Web Content Accessibility Guidelines (WCAG)*.
ANDI	*Accessible Name & Description Inspector.* A browser-based inspection tool for verifying accessible names, roles, and states of HTML elements.
API	*Application Programming Interface.* A set of routines or tools that allows different software components to communicate and share functionality.
Assistive Technology (AT)	Hardware or software (e.g., screen readers, adaptive keyboards, voice control) that helps people with disabilities access digital content.
Black-Box Testing	A testing method that evaluates external functionality and user experience without examining the internal code or structure.
Bug Triage	The process of reviewing, classifying, and prioritizing reported defects based on severity and user impact.
Keyboard Accessibility	The ability to operate all interactive elements via keyboard input alone, without the need for mouse or touch.
NVDA / VoiceOver	Common screen-reader applications used by visually impaired users (NVDA – Windows, VoiceOver – macOS / iOS).
P001–P004	Persona identifiers used in this project: P001 – Visual Impairment, P002 – Active Mobile User, P003 – Motor Difficulties, P004 – Inexperienced User.
Pointer Gestures	Touch- or mouse-based actions such as drag, swipe, or pinch used for interaction; must have accessible alternatives.
Screen Reader	Assistive software that reads aloud on-screen text and describes interface elements for blind or low-vision users.
Target Size	Minimum physical size and spacing of interactive elements (WCAG 2.5.5) to ensure they can be selected accurately.
Test Case	A documented sequence of steps, expected results, and conditions used to verify a specific requirement or scenario.
Test Log	A chronological record of test executions, outcomes, and observed behaviors.
Usability Testing	Evaluation of how easy and intuitive a system is for its intended users; accessibility testing is a subset of this practice.
WAVE	*Web Accessibility Evaluation Tool* used to automatically detect accessibility and WCAG compliance issues.
WCAG	*Web Content Accessibility Guidelines (2.1)* – international standards developed by the W3C for making web content accessible to everyone.
W3C	*World Wide Web Consortium* – the organization responsible for developing web standards, including WCAG and HTML.

References

Hambling B. (2010). *Software Testing: An ISTQB–ISEB Foundation Guide: Second Edition*. British Informatics Society Limited

Hamilton, T. (n.d.-a). *Accessibility Testing Tutorial: What is, Tools & Examples*. Retrieved 12 October 2021, from https://www.guru99.com/accessibility-testing.html

Hamilton, T. (n.d.-b). *Software Testing Methodologies: Learn QA Models*. Retrieved 12 October 2021, from https://www.guru99.com/testing-methodology.html

WCAG_2.1_Checklist.pdf. (n.d.). Retrieved 12 October 2021, from https://kma.global/wp-content/uploads/2019/07/WCAG_2.1_Checklist.pdf

Hambling, B. (2010). *Software Testing: An ISTQB–BCS Certified Tester Foundation Guide* (2nd ed.). British Informatics Society Limited.

World Wide Web Consortium (W3C). (2018). *Web Content Accessibility Guidelines (WCAG) 2.1*. Retrieved from https://www.w3.org/TR/WCAG21/